

UJAN DEY

Physics Undergraduate | Machine Learning | Continual Learning | Neuromorphic Computing

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EDUCATION

National Institute of Technology Agartala

2025 - 2030 (Expected)

Integrated B.S.-M.S. in Physics | Relevant Coursework: Probability & Statistics, Computer Science, Mathematical Physics

RESEARCH INTERESTS

Continual Learning • Neuromorphic Computing • Spiking Neural Networks • Efficient Machine Learning • Deep Learning

RESEARCH EXPERIENCE / PROJECT

MNEMA - Neuromorphic Continual Learning Architecture

Jun 2026 - Present

Student Research Project | github.com/ujandey/COGNX_2

- Developed and evaluated a modular continual-learning architecture combining temporal event encoding, sparse associative memory, adaptive spiking dynamics, neuromodulatory control and offline consolidation.
- Evaluated on **full Split-MNIST Class-Incremental Learning across 10 seeds**, achieving **77.12 ± 0.87% final accuracy** with **6.53 ± 1.41 pp forgetting** under 10-class task-agnostic inference.
- Outperformed a strict **64 KiB replay baseline** in accuracy (**77.12% vs 67.59%**) and forgetting (**6.53 vs 39.30 pp**); DER++ achieved higher final accuracy at 89.39%.
- Built a reproducible multi-seed evaluation pipeline with retention matrices, memory accounting, state-isolated testing and accuracy-memory trade-off analysis.

SELECTED PROJECT

FIFA World Cup Outcome Prediction Model

2026

- Built a probabilistic tournament-prediction pipeline using **Elo ratings, Logistic Regression/XGBoost and 100,000 Monte Carlo simulations** to estimate championship probabilities.

TECHNICAL SKILLS

Programming & Scientific Computing: Python, NumPy, Pandas, vectorized numerical computing, data preprocessing

Machine Learning: regression, classification, logistic regression, regularization, model evaluation, feature scaling, gradient-based optimization

Deep Learning: TensorFlow, PyTorch (learning), feedforward neural networks, activations, loss functions, backpropagation fundamentals

Continual / Neuromorphic Learning: catastrophic forgetting, Class-IL, experience replay, EWC, DER++, spiking neural networks, sparse representations, associative memory

Experimentation & Tools: Matplotlib, Monte Carlo simulation, multi-seed evaluation, retention matrices, Git, GitHub, GitHub Actions, Linux/CLI basics

ADDITIONAL

Self-directed study in Machine Learning and Deep Learning through online coursework, technical documentation and implementation-based learning.